

## Human Review: Ray's $B_{p,q}$ embedding conjecture

Original automatic proof: `solution_packet.pdf`

Checked date: 13 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

## Human Comments

The proof appears to give a correct answer to Ray's conjecture on when

$$(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$$

embeds linearly isometrically into  $S_\infty$ . The positive cases  $(2,2)$ ,  $(1,2)$ , and  $(2,1)$  are handled by explicit  $2 \times 2$  matrix embeddings, hence by finite-rank embeddings into  $S_\infty$ .

The negative cases use Ray's existing results to exclude  $(1,1)$  and the region  $\max\{p,q\} > 2$ . The main new part is the exclusion of the remaining fractional region, where one exponent lies strictly between 1 and 2.

## Axis-Curve Lemma

The key point is the axis-curve lemma. If an isometry existed and

$$\Phi(z_1, z_2) = z_1 T + z_2 S$$

with  $T, S \in S_\infty$ , then along the first coordinate axis one would have

$$\|T + tS\| = \|(1, t)\|_{B_{p,q}}.$$

Since  $T$  is compact and  $\|T\| = 1$ , the norm-one spectral cluster of

$$(T + tS)^*(T + tS)$$

near  $t = 0$  is finite-dimensional and isolated. Analytic perturbation theory then says that the relevant eigenvalues are finitely many real analytic functions of  $t$ . Consequently, if

$$\|T + tS\|^2 > 1$$

for small positive  $t$ , then

$$\|T + tS\|^2 - 1 = ct^m + o(t^m)$$

for some integer  $m \geq 1$  and some  $c > 0$ .

Thus the compact-operator norm curve cannot have a first nonzero term with a non-integer exponent. This is exactly what rules out the fractional exponents.

## Axis Asymptotic

For the  $B_{p,q}$ -norm, put

$$\lambda(t) = \|(1, t)\|_{B_{p,q}}.$$

By definition,

$$\lambda(t)^{-p} + t^q \lambda(t)^{-q} = 1.$$

Since  $\lambda(t) \rightarrow 1$ , write

$$\lambda(t) = 1 + u(t), \quad u(t) \rightarrow 0.$$

Then

$$(1 + u)^{-p} + t^q (1 + u)^{-q} = 1.$$

Equivalently,

$$1 - (1 + u)^{-p} = t^q (1 + u)^{-q}.$$

The left-hand side satisfies

$$1 - (1 + u)^{-p} = pu + O(u^2),$$

while  $(1 + u)^{-q}$  remains bounded above and below by positive constants. Therefore the equation first gives

$$u = O(t^q).$$

Using this in the expansion,

$$(1 + u)^{-p} = 1 - pu + O(u^2), \quad (1 + u)^{-q} = 1 + O(u),$$

we get

$$-pu + t^q + O(u^2) + O(t^q u) = 0.$$

Since  $u = O(t^q)$ , both error terms are  $o(t^q)$ , and hence

$$u(t) = \frac{1}{p} t^q + o(t^q).$$

It follows that

$$\|(1, t)\|_{B_{p,q}}^2 - 1 = \lambda(t)^2 - 1 = \frac{2}{p} t^q + o(t^q).$$

If  $1 < q < 2$ , this contradicts the integer-power conclusion of the axis-curve lemma. The same argument along the other coordinate axis, using  $(t, 1)$ , excludes  $1 < p < 2$ .

## Literature and Follow-Up

The conjecture is present in the journal version of Ray's paper, on printed page 445 of the BLMS article. A preliminary literature check found later related work on Schatten-class embeddability, in particular work of Chattopadhyay–Hong–Pal–Pradhan–Ray, but did not find a later resolution of the full off-diagonal  $B_{p,q}$  conjecture.

The result seems interesting. I suggest reaching out to Ray to ask whether the conjecture has since been resolved elsewhere, whether the argument is known to him, and how he would assess the significance of the proof.

## Review Outcome

Archive this packet as an apparently correct full solution to Ray's  $B_{p,q}$  isometric-embedding conjecture, pending author feedback and a more thorough final literature check. The proof should make explicit the  $u = O(t^q)$  step in the axis asymptotic, since this is currently implicit.